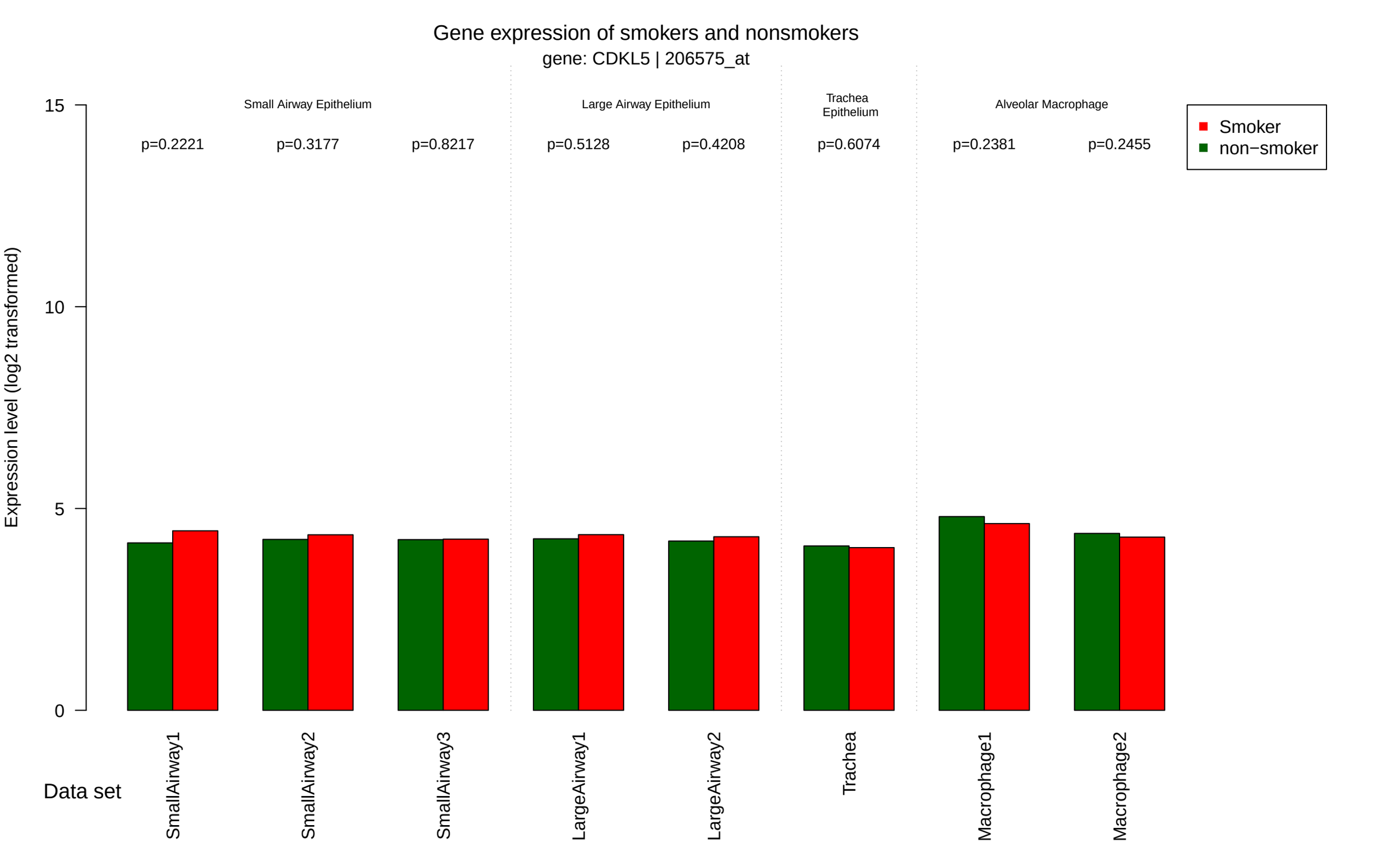
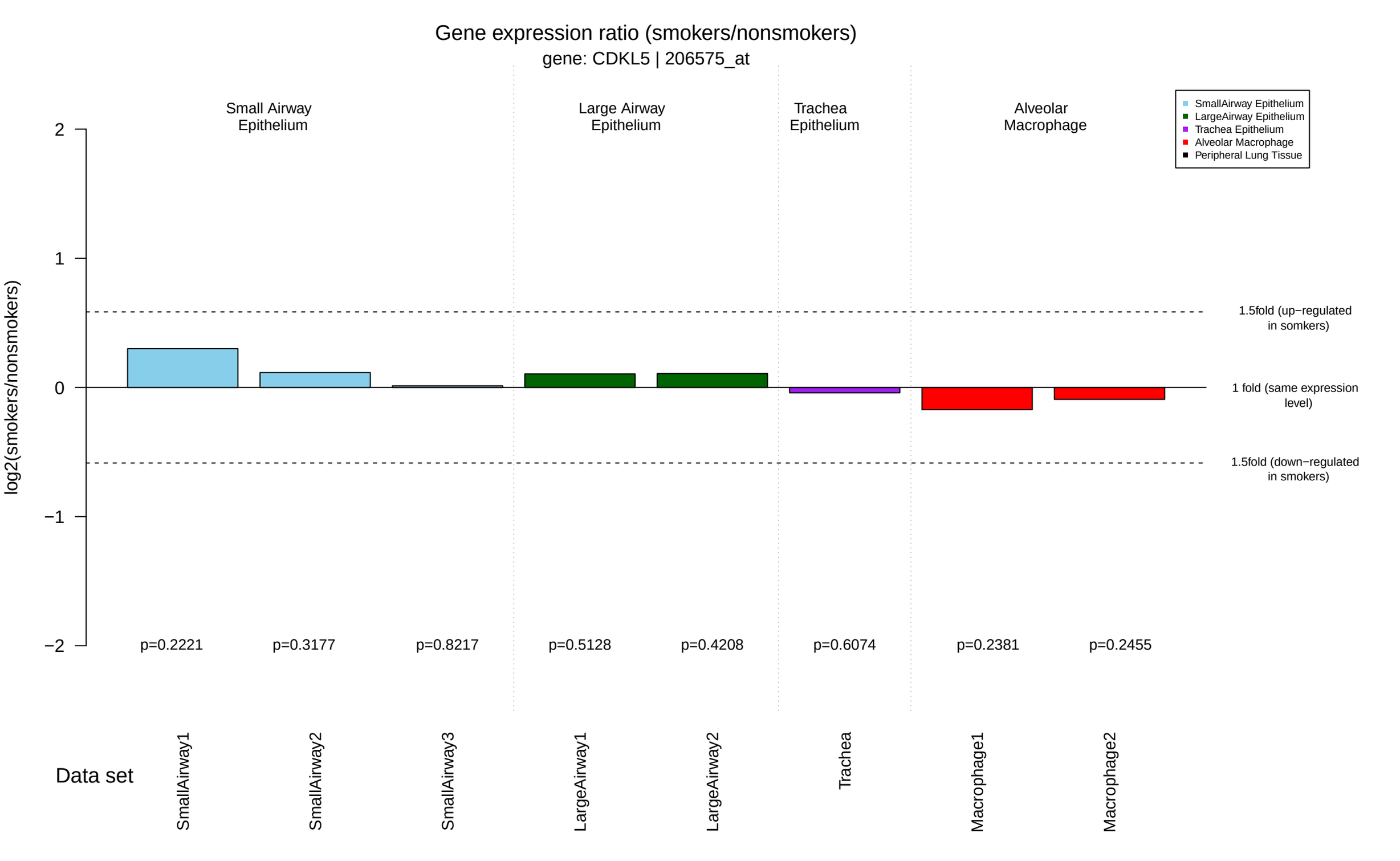
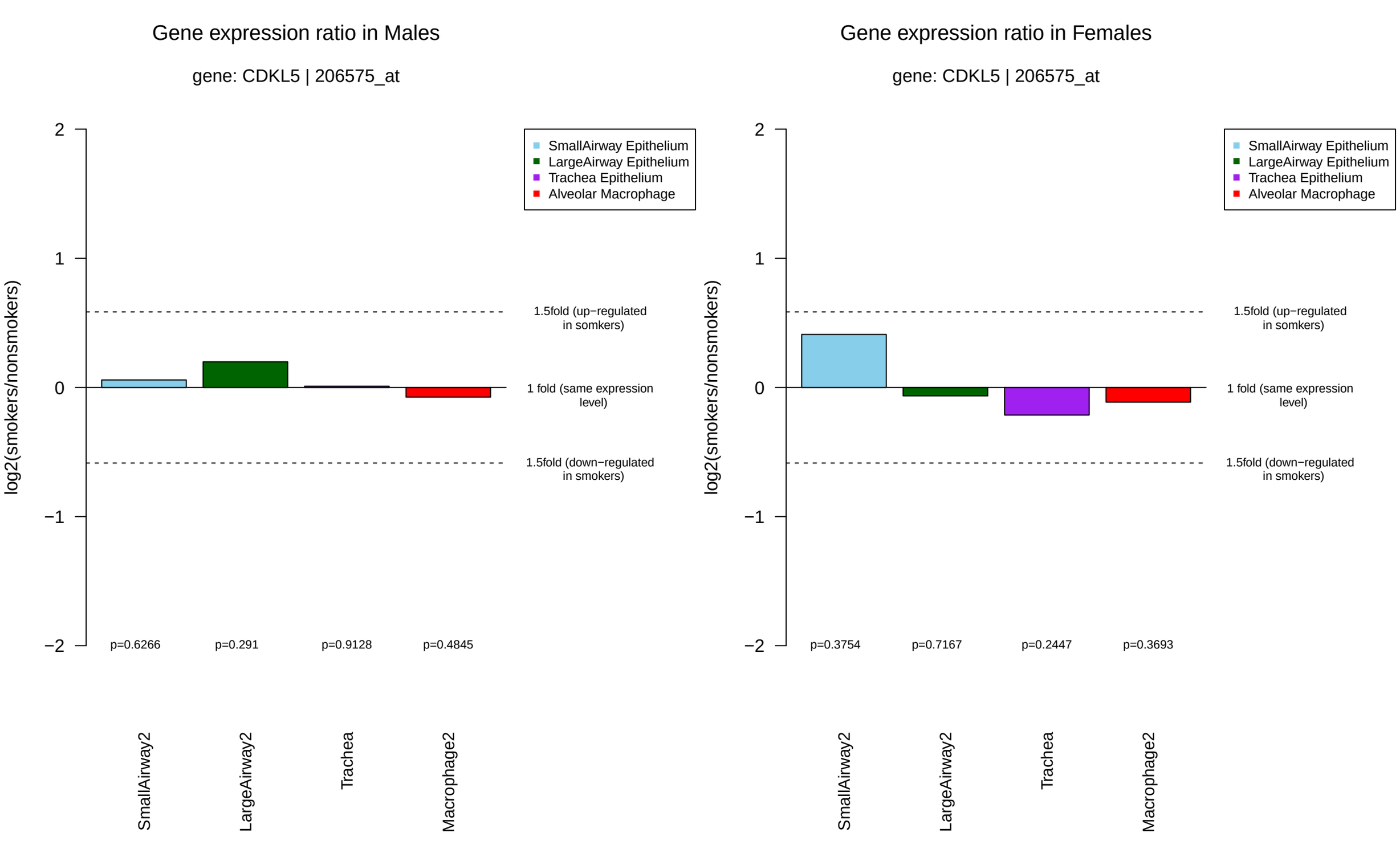
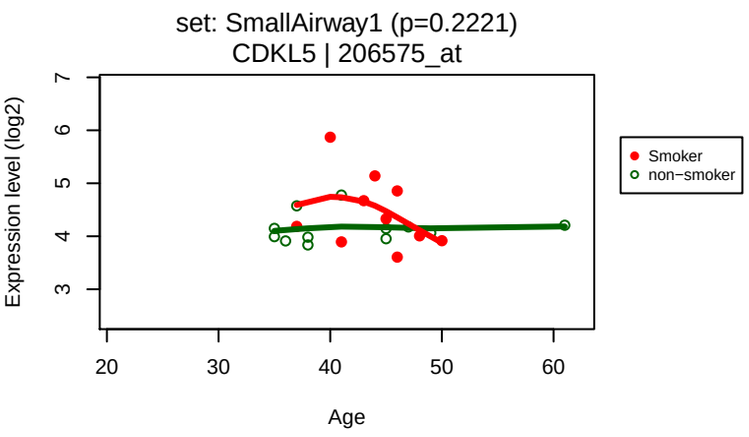
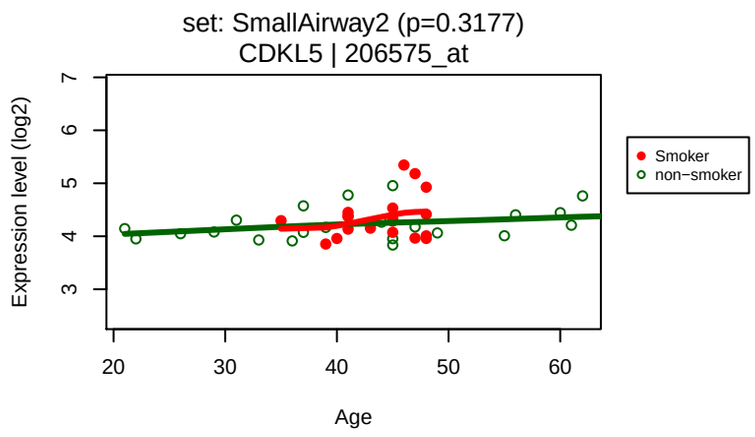
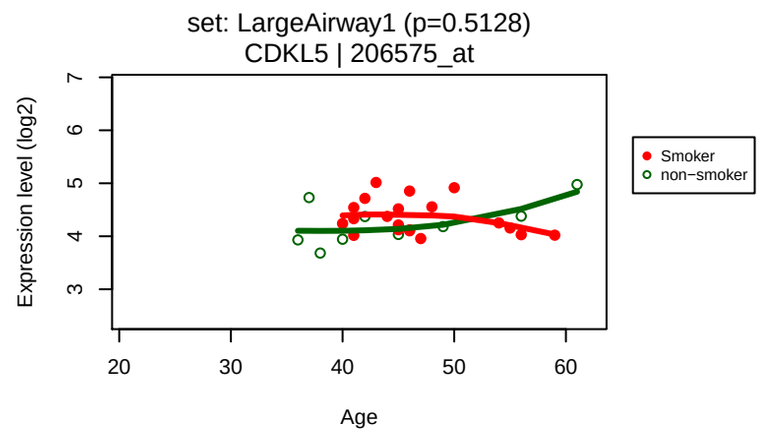

Gene expression of smokers and nonsmokers
gene: CDKL5 | 206575_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2221
p=0.3177
p=0.8217
p=0.5128
p=0.4208
p=0.6074
p=0.2381
p=0.2455
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

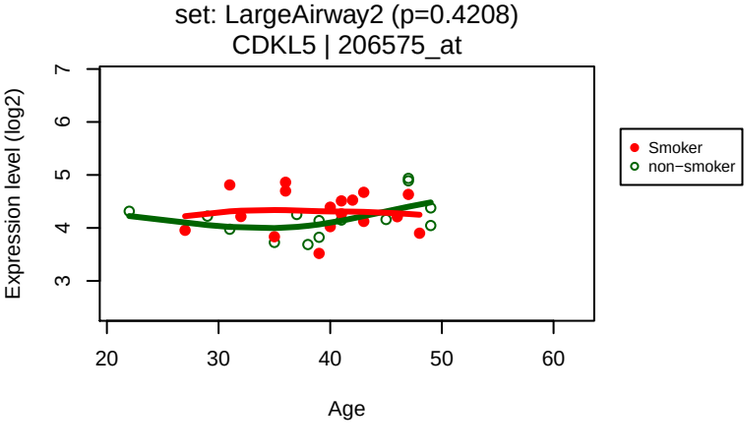

Gene expression ratio (smokers/nonsmokers)
gene: CDKL5 | 206575_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2221
p=0.3177
p=0.8217
p=0.5128
p=0.4208
p=0.6074
p=0.2381
p=0.2455
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

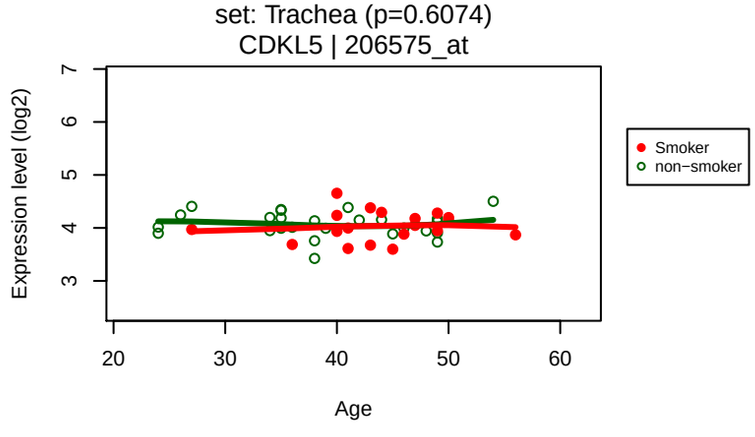

Gene expression ratio in Males
gene: CDKL5 | 206575_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6266
p=0.291
p=0.9128
p=0.4845
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDKL5 | 206575_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3754
p=0.7167
p=0.2447
p=0.3693
SmallAirway2
LargeAirway2
Trachea
Macrophage2

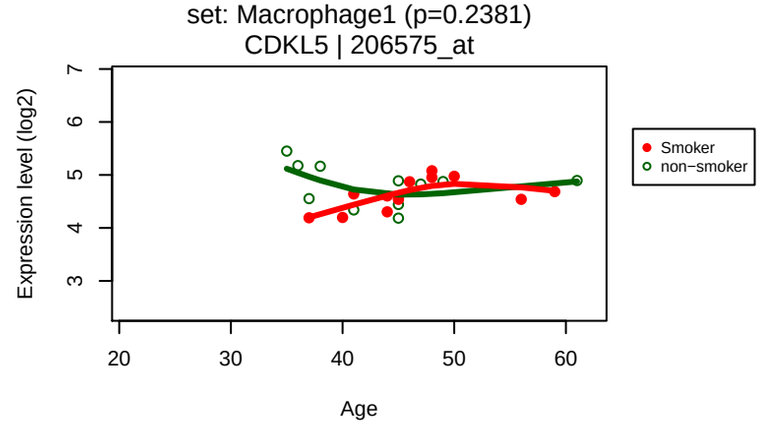

set: SmallAirway1 (p=0.2221)
CDKL5 | 206575_at
Expression level (log2)
Age
Smoker
non−smoker

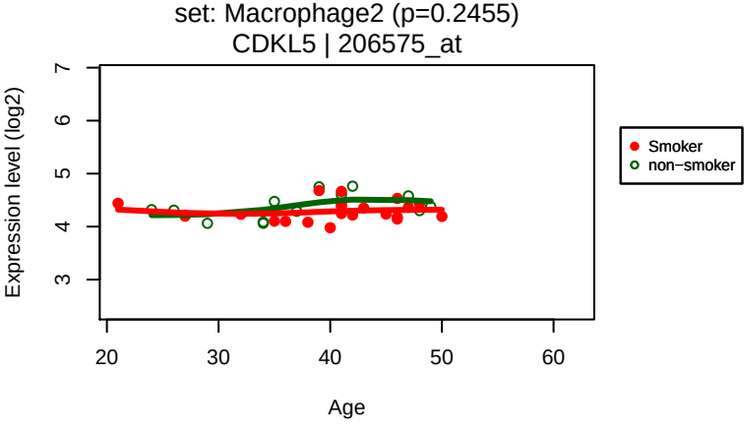

set: SmallAirway2 (p=0.3177)
CDKL5 | 206575_at
Expression level (log2)
Age
Smoker
non−smoker

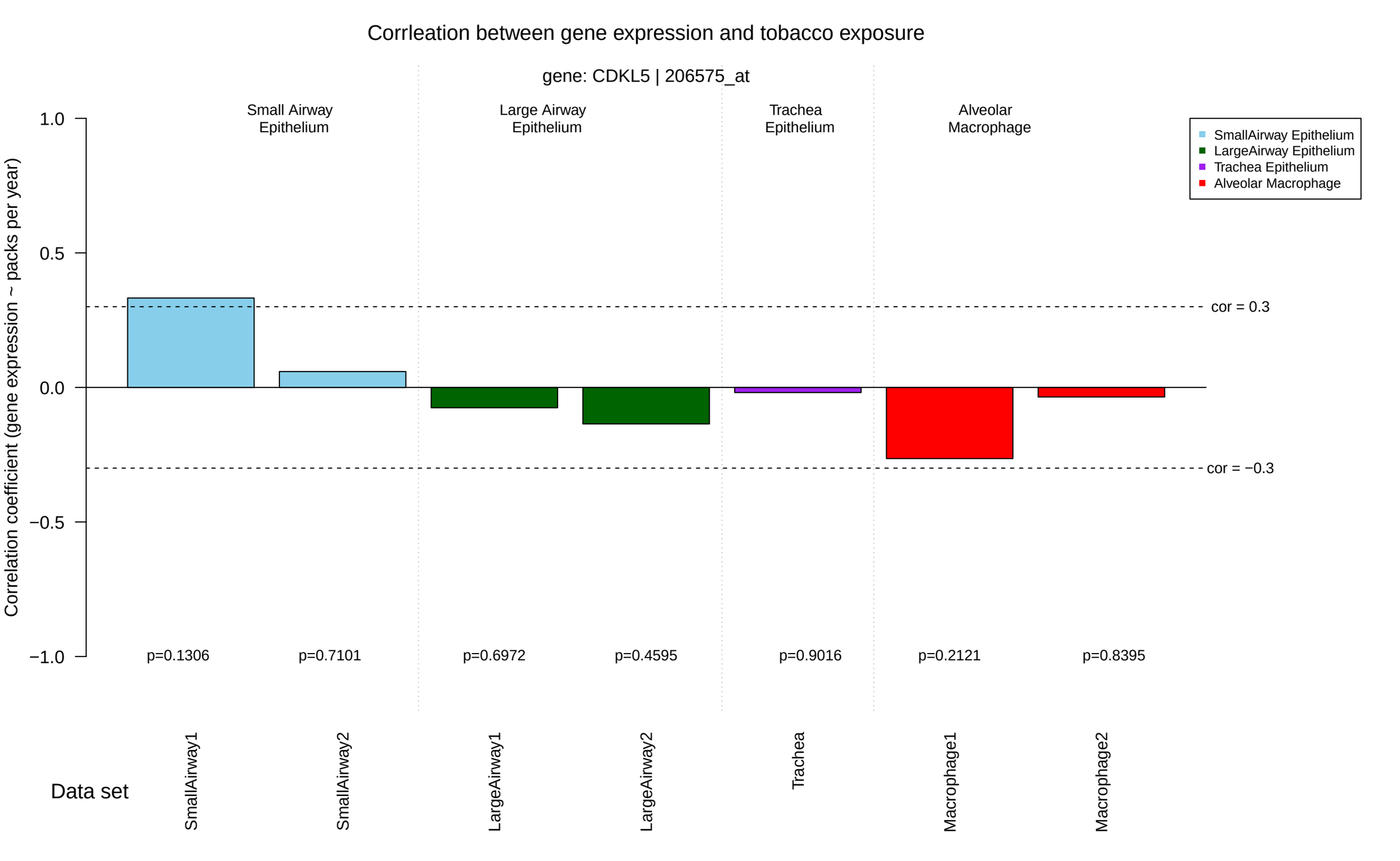

set: LargeAirway1 (p=0.5128)
CDKL5 | 206575_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4208)
CDKL5 | 206575_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6074)
CDKL5 | 206575_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2381)
CDKL5 | 206575_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2455)
CDKL5 | 206575_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDKL5 | 206575_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1306
p=0.7101
p=0.6972
p=0.4595
p=0.9016
p=0.2121
p=0.8395
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2